MILITARY STANDARD

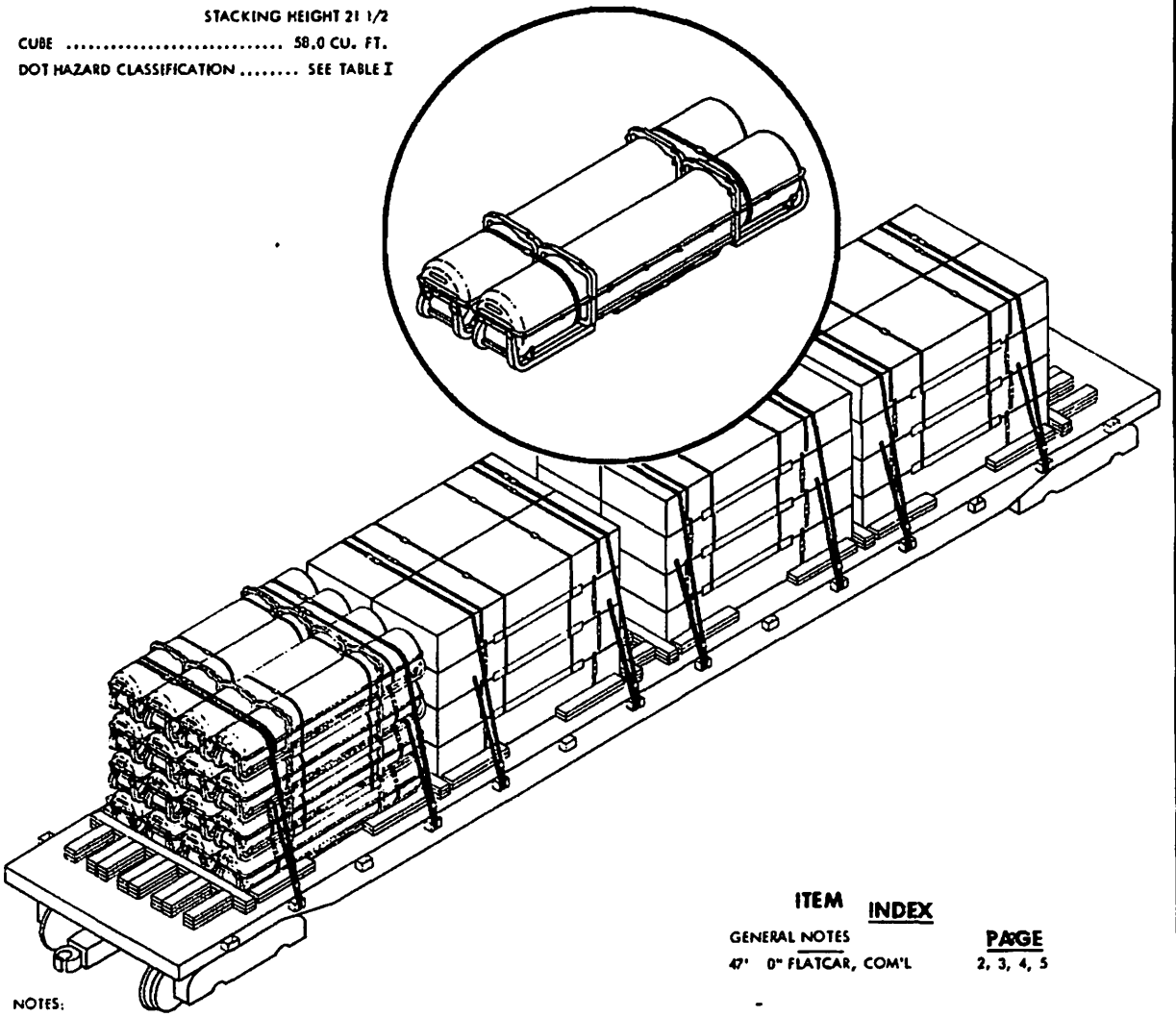
MIL-STD-1325-193
(NAVY)

RAILCAR LOADING
TORPEDO MK 44 AND MK 46 ASSEMBLIES IN CONTAINER MK 197 MOD 1

5 JANUARY 1978

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/241
WEIGHT (APPROXIMATE) ..... SEE TABLE I
DIMENSIONS ..... 114 L X 40 W X 22 H
STACKING HEIGHT 21 1/2
CUBE ..... 58.0 CU. FT.
DOT HAZARD CLASSIFICATION ..... SEE TABLE I



ITEM INDEX

GENERAL NOTES	PAGE
47' 0" FLATCAR, COM'L	2, 3, 4, 5

NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCK-LOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HD-BK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

APPROVED BY BUREAU OF EXPLOSIVES

A. F. Grammuck 12/14/77
SIGNATURE SUPERVISOR, MILITARY AND INTERMODEL SERVICES DATE

M S Gray, NWHC 1/3/78
TECHNICAL DIRECTION AGENT (TDA) DATE

J C Kelly 1/3/78
SEA SYSCOM, BY DIRECTION DATE

ORIGINATOR J F C Burk 1/3/78
DATE

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:04Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. LOADING PLANS SHOWN ARE FOR FLATCARS WITH A PLATFORM 47 FEET 0 INCHES LONG AND A MINIMUM OF 9 FEET 0 INCHES WIDE WITH STAKE POCKETS SPACED 42 INCHES ON CENTERS. FLATCARS OTHER THAN 47 FEET 0 INCHES LONG MAY BE USED PROVIDED THE LOADING, BLOCKING, AND STRAPPING PROCEDURES SHOWN IN THIS STANDARD ARE FOLLOWED. FLATCARS WITH STEEL FLOOR ENDS AND/OR EXPOSED STEEL BOLSTERS THAT INTERFERE WITH NAILING SHALL NOT BE USED.
3. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
4. THE LOAD CONSISTS OF MK 44 AND/OR MK 46 TORPEDO ASSEMBLIES IN CONTAINER MK 197 MOD 1 UNITIZED IN ACCORDANCE WITH WR-54/241.
5. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
6. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
7. ALL STRAP JOINTS, EXCEPT FOR RESTRAINER RETAINING STRAP, PIECE 2, SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
8. APPLICABLE MATERIAL SPECIFICATIONS:
   DUNNAGE LUMBER - FED. SPEC MM-L-751
   NAILS - FED. SPEC FF-N-105
   STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, FINISH A, B, OR C.
   SEALS - FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
9. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS IN AN ACCESSIBLE AREA, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL I).

47 FT 0 IN FLATCAR, COMMERCIAL

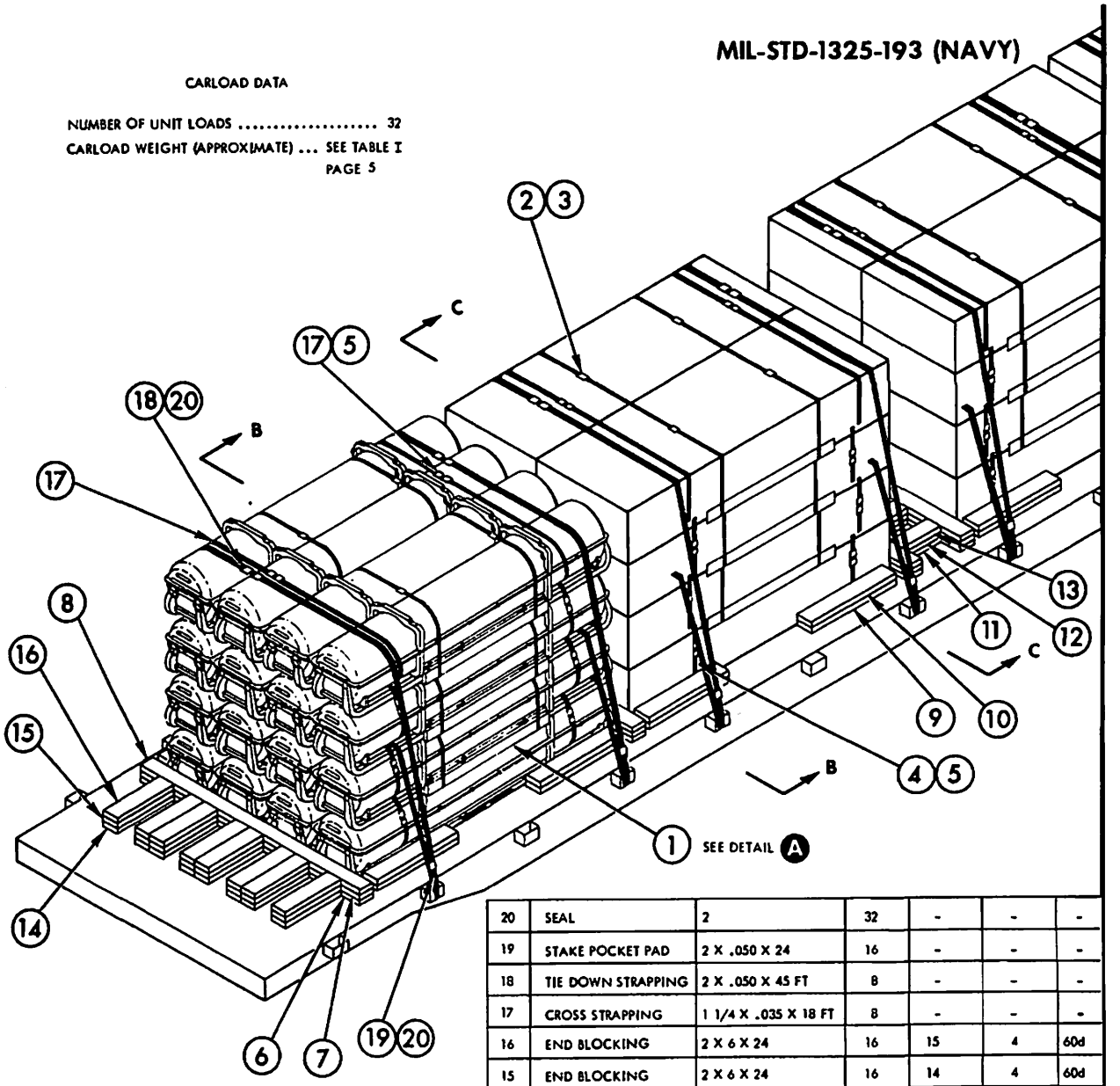
1. THE CARLOAD CONSISTS OF 32 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED ON COMMERCIAL FLATCARS THE SAME PROCEDURES AND METHODS OF BLOCKING AND STRAPPING ARE APPLICABLE. WHEN A STACK CONSISTS OF AN ODD NUMBER OF UNIT LOADS, A UNIT LOAD OF EMPTY CONTAINERS MUST BE USED TO FILL OUT THE TOP LAYER OF THE STACK OR IF FLOOR SPACE PERMITS THE ODD UNIT LOAD MAY BE BLOCKED AND STRAPPED ON THE CAR FLOOR.
3. EACH UPPER UNIT LOAD MUST BE STRAPPED TO THE UNIT LOAD BENEATH WITH TWO BANDS OF STEEL STRAPPING (PIECE 4) POSITIONED AS SHOWN (SEE SECTION B-B PAGE 5) AND SECURED WITH TWO DOUBLE CRIMPED SEALS (PIECE 5) PER BAND. IN ADDITION, TO PREVENT LONGITUDINAL SHIFTING OF UPPER LAYERS, EACH UPPER UNIT LOAD MUST HAVE A RESTRAINER (PIECE 1) POSITIONED EACH SIDE AND HELD IN PLACE BY STRAPPING (PIECE 2) AS SHOWN (SEE DETAIL A PAGE 4). THIS RESTRAINER IS REQUIRED ON FLATCAR LOADS TO SUPPLEMENT THE LONGITUDINAL RESTRAINT AFFORDED BY THE STACKING RINGS.
4. STRAP THE TOP LAYER OF EACH STACK OF UNIT LOADS TOGETHER WITH TWO BANDS OF STEEL STRAPPING (PIECE 17) POSITIONED AS SHOWN (SEE SECTION B-B PAGE 5), AND SECURE WITH TWO DOUBLE CRIMPED SEALS (PIECE 5) PER BAND.
5. STRAP THE LOAD TO THE CAR WITH TWO BANDS OF STEEL TIE DOWN STRAPPING (PIECE 18) OVER EACH STACK, POSITIONED AS SHOWN (SEE SECTION C-C PAGE 5). TENSION AND SECURE THE TIE DOWN STRAPPING TO THE CAR AS FOLLOWS:
   A. RUN TIE DOWN STRAPPING UP THROUGH THE STAKE POCKET ON ONE SIDE OF THE CAR, THREAD THROUGH THE STACK OVER THE SKIDS OF THE THIRD LAYER UNIT LOADS, AND DOWN THROUGH THE STAKE POCKET ON THE OTHER SIDE OF THE CAR.
   B. THREAD ONE SEAL ON EACH END OF THE TIE DOWN STRAPPING, POSITION STAKE POCKET PADS (PIECE 19) IN STAKE POCKETS AS SHOWN, RUN TIE DOWN STRAPPING UP EACH SIDE OF THE STACK AND HOLD STAKE POCKET PADS WITH SEALS.
   C. TENSION TIE DOWN STRAPPING ON TOP OF THE STACK AND SEAL WITH TWO DOUBLE CRIMPED SEALS.
   D. PUT SINGLE CRIMP IN EACH OF THE STAKE POCKET PAD SEALS.

Source: https://assist.dla.mil -- Downloaded: 202640-24T00:04Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:04Z
Check the source to verify that this is the current version before use.

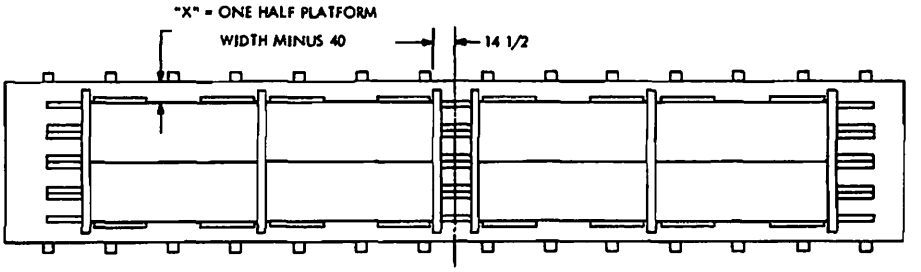
CARLOAD DATA

NUMBER OF UNIT LOADS 32
CARLOAD WEIGHT (APPROXIMATE) ... SEE TABLE I
PAGE 5



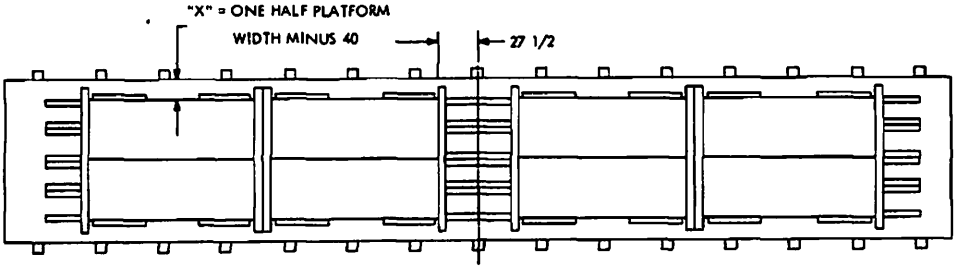
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
20	SEAL	2	32	-	-	-
19	STAKE POCKET PAD	2 X .050 X 24	16	-	-	-
18	TIE DOWN STRAPPING	2 X .050 X 45 FT	8	-	-	-
17	CROSS STRAPPING	1 1/4 X .035 X 18 FT	8	-	-	-
16	END BLOCKING	2 X 6 X 24	16	15	4	60d
15	END BLOCKING	2 X 6 X 24	16	14	4	60d
14	END BLOCKING	2 X 6 X 24	16	CAR FLOOR	4	30d
13	INTERMEDIATE BLOCKING	2 X 6 X 18	8	12	3	60d
12	INTERMEDIATE BLOCKING	2 X 6 X 18	8	11	3	60d
11	INTERMEDIATE BLOCKING	2 X 6 X 18	8	CAR FLOOR	3	30d
10	SIDE BLOCKING	2 X 6 X36	16	9	6	30d
9	SIDE BLOCKING	2 X 6 X 36	16	CAR FLOOR	6	30d
8	CROSS MEMBER	2 X 6 X 96	6	7	1 EVERY 8 INCHES	60d
7	CROSS MEMBER	2 X 6 X 96	6	6	1 EVERY 8 INCHES	60d
6	CROSS MEMBER	2 X 6 X 96	6	CAR FLOOR	1 EVERY 8 INCHES	30d
5	SEAL	1 1/4	112	-	-	-
4	VERTICAL STRAPPING	1 1/4 X .035 X 11 FT	48	-	-	-
3	SEAL	3/4	48	-	-	-
2	RESTRAINER RETAINING STRAP	3/4 X .035 X 12 FT	48	-	-	-
1	LONGITUDINAL RESTRAINER	2 X 4 X CUT TO SUIT	48	SEE DETAIL A PAGE 4		

LIST OF MATERIALS & NAILING DATA



LOADING PLAN

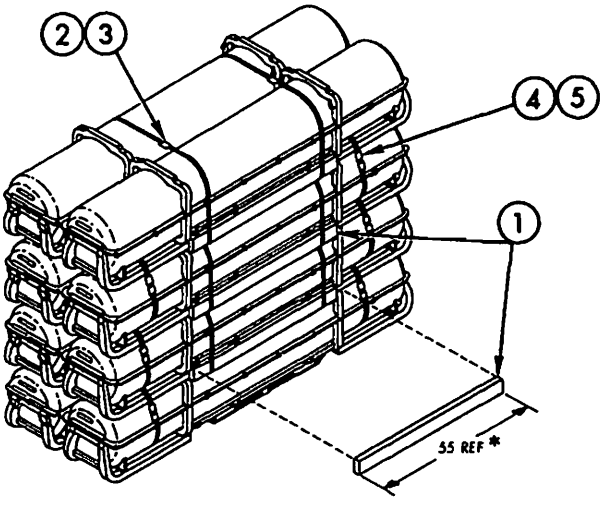
(FLATCARS WITH 47 FT. TO 50 FT. 6 IN. LENGTH
AND 14 STAKE POCKETS EACH SIDE)



LOADING PLAN

(FLATCARS WITH 52 FT. TO 53 FT. 6 IN. LENGTH
AND 15 STAKE POCKETS EACH SIDE)

THE RESTRAINERS, PIECE 1, PREVENT LONGITUDINAL SHIFTING OF UPPER UNIT LOADS WITH RESPECT TO LOWER UNIT LOADS AND ARE HELD IN POSITION BY STRAPPING, PIECE 2. THEY SHOULD BE FIELD POSITIONED AND CUT TO SUIT. TWO 10d NAILS SHOULD BE DRIVEN HALF WAY INTO THE EDGE (1 1/2 INCH SIDE) OF THE 2 X 4 RESTRAINER (APPROXIMATELY 15 INCHES FROM ONE END AND 11 INCHES FROM THE OTHER END) SO THAT WHEN RESTRAINERS ARE POSITIONED THE HEADS OF THE NAILS PROTRUDE THROUGH PAD EYES ON THE CONTAINER SIDE. THE NAILS MAY THEN BE BENT SLIGHTLY OUTWARD TO HOLD THE RESTRAINER IN POSITION UNTIL PREPOSITIONED STRAPS (PIECE 2) ARE TENSIONED AND SEALED.

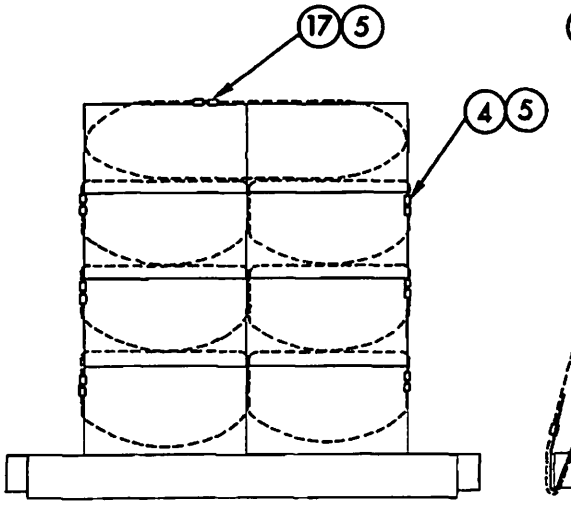


* CUT TO SUIT TO ASSURE TIGHT FIT

DETAIL A

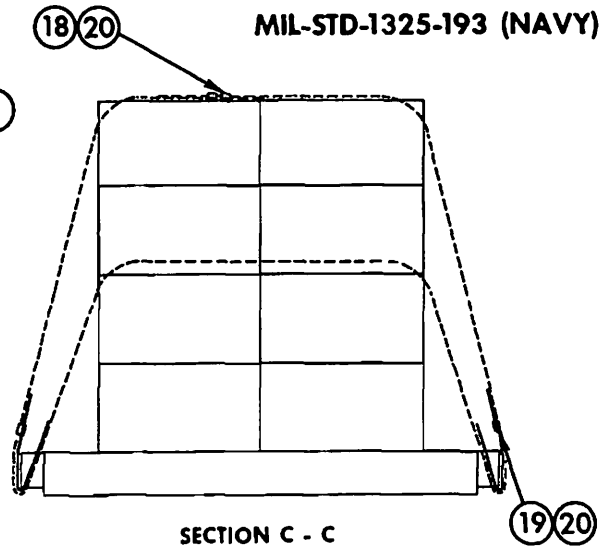
SHOWING RESTRAINER INSTALLATION
AND VERTICAL STRAPPING

Source: https://assist.dla.mil -- Downloaded: 202640-24T00:04Z
Check the source to verify that this is the current version before use.



SECTION B - B

SHOWING VERTICAL STRAPPING AND CROSS STRAPPING
(TIE DOWN STRAPPING AND FLOOR BLOCKING OMITTED FOR CLARITY)



SECTION C - C

SHOWING DOUBLE THREADED TIE DOWN STRAPPING WITH STAKE POCKET PADS
(ALL OTHER STRAPPING AND FLOOR BLOCKING OMITTED FOR CLARITY)

TABLE I

DODIC / NALC	ITEM		UNIT LOAD WEIGHT (LBS., APPROX.)	DOT HAZARD CLASSIFICATION	CARLOAD WEIGHT* (LBS., APPROX.)
T018	MAIN ASSEMBLY, TORPEDO MK 44 MOD 1		1204	INERT	39691
T026	MAIN ASSEMBLY, TORPEDO MK 46 MOD 1	W/O FUEL	1318	INERT	43339
		W/FUEL	1494	N/A	48971
T029	MAIN ASSEMBLY, TORPEDO MK 46 MOD 2	W/O FUEL	1318	INERT	43339
		W/FUEL	1494	N/A	48971
T031	MAIN ASSEMBLY, TORPEDO MK 46 MOD 1 (PHASE 2)	W/O FUEL	1318	INERT	43339
		W/FUEL	1494	N/A	48971
T042	DUMMY TORPEDO MK 44 (WOOD)		1508	INERT	49419
T043	DUMMY TORPEDO MK 44 (STEEL)		1508	INERT	49419
T044	DUMMY TORPEDO MK 46 MOD 1 (WOOD)		1686	INERT	55115
T045	DUMMY TORPEDO MK 46 MOD 1 (STEEL)		1686	INERT	55115
1510	TORPEDO MK 46 MOD 1, PHASE 2, WARSHOT COMPLETE ASSEMBLY, W/O LAUNCH ASSEMBLY		1682	A	54987
1512	TORPEDO MK 46 MOD 1, PHASE 2, WARSHOT, COMPLETE ASSEMBLY, TUBE LAUNCHED		1686	A	55115
1536	TORPEDO MK 44 MOD 1, WARSHOT, COMPLETE ASSEMBLY, TUBE LAUNCHED		1536	A	50315
1537	TORPEDO MK 44 MOD 1, EXERCISE, COMPLETE ASSEMBLY, TUBE LAUNCHED		1498	AMMO NON EXPL	49099
1540	TORPEDO MK 46 MOD 1, WARSHOT, COMPLETE ASSEMBLY, TUBE LAUNCHED		1686	A	55115
1573	TORPEDO MK 46 MOD 1, EXERCISE, COMPLETE ASSEMBLY, TUBE LAUNCHED	W/LONG TANK	1686	N/A	55115
		W/SHORT TANK	1612	N/A	52747
1592	TORPEDO MK 46 MOD 2, WARSHOT, COMPLETE ASSEMBLY, TUBE LAUNCHED		1686	A	55115
1593	TORPEDO MK 46 MOD 2, EXERCISE, COMPLETE ASSEMBLY, TUBE LAUNCHED	W/LONG TANK	1686	N/A	55115
		W/SHORT TANK	1612	N/A	52747

*INCLUDES 1163 LBS. (APPROX.) DUNNAGE WEIGHT

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:04Z
Check the source to verify that this is the current version before use.